

CHE 321 - Exam 2 - Sample 2

examProblem_fun.m

```
1 function h=examProblem_fun(v,t)
2 Fa = t(1);
3 Fb = t(2);
4 Fc = t(3);
5 Fd = t(4);
6 T = t(5);
7 Ta = t(6);
8
9 Cpa = 10;
10 Cpb = 10;
11 Cpc = 30;
12 Cpd = 20;
13
14 E2 = 12000;
15 y = 1;
16 R = 1.9872;
17 E1 = 8000;
18 T0 = 300;
19 Ct0 = 0.2;
20 Cpco = 10;
21 m = 50;
22 DH1b = -15000;
23 DH2a = -10000;
24 kc = 0.5;
25 Ua = 80;
26
27 Ft = Fa+Fb+Fc+Fd;
28
29 k1a = 40*exp((E1/R)*(1/300-1/T));
30 K2c = 2*exp(((E2/R*(1/300-1/T)))));
31 sumFiCPI = Cpa*Fa+Cpb*Fb+Cpc*Fc+Cpd*Fd;
32
33 Ca = Ct0*(Fa/Ft)*(T0/T)*y;
34 Cb = Ct0*(Fb/Ft)*(T0/T)*y;
35 Cc = Ct0*(Fc/Ft)*(T0/T)*y;
36
37 r2c = -K2c*Ca*Cc;
38 r1a = -k1a*Ca*Cb^2;
39 r1b = 2*r1a;
40 rb = r1b;
41 r2a = r2c;
42
43 r1c = -r1a;
44 rc = r1c + r2c;
45 r2d = -2*r2c;
46 ra = r1a+r2a;
47 rd = r2d;
48
49 Qg = r1b*DH1b+r2a*DH2a;
50 Qr = Ua*(T-Ta);
51
52 h1=ra;
53 h2=rb;
54 h3=rc-kc*Cc;
55 h4=rd;
56 h5=(Qg-Qr)/sumFiCPI;
57 h6=Ua*(Ta-T)/m/Cpco;
58
59 h = [h1 h2 h3 h4 h5 h6]';
```

examProblem_simi.m

```
1  clc
2  clear all;
3
4  t0 = [5 10 0 0 300 511.9855];
5
6  vspan = [0 10];
7
8  [v,t] = ode45(@examProblem_fun,vspan,t0);
9
10 Fa = t(:,1);
11 Fb = t(:,2);
12 Fc = t(:,3);
13 Fd = t(:,4);
14 T = t(:,5);
15
16 Ta = t(:,6);
17
18 figure(1)
19 plot(v,Fa)
20 hold on
21 plot(v,Fb)
22 plot(v,Fc)
23 plot(v,Fd)
24 legend('Fa','Fb','Fc','Fd')
25 figure(2)
26 plot(v,T)
27 hold on
28 plot(v,Ta)
29 legend('T','Ta')
30
31 T0=300;
32 y=1;
33 Ft=Fa+Fb+Fc+Fd;
34 Ct0=0.2;
35
36 Ca=Ct0.*(Fa./Ft).*(T0./T).*y;
37 Cb=Ct0.*(Fb./Ft).*(T0./T).*y;
38 Cc=Ct0.*(Fc./Ft).*(T0./T).*y;
39 Cd=Ct0.*(Fd./Ft).*(T0./T).*y;
40
41 figure(3)
42 plot(v,Ca)
43 hold on
44 plot(v,Cb)
45 plot(v,Cc)
46 plot(v,Cd)
47 legend('Ca','Cb','Cc','Cd')
```